BRUTE FORCE SSH ATTACK DETECTION & MONITORING

Security Operations Center (SOC) Project Report

Tool: Splunk Enterprise SIEM
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Platform: Ubuntu Linux + Splunk Enterprise

1. Executive Summary

This report documents the design, implementation, and results of a Brute Force SSH Attack Detection project using Splunk Enterprise as the Security Information and Event Management (SIEM) platform. The project was carried out in a controlled local lab environment on Ubuntu Linux.

The project involved three core phases:

- Attack Simulation: Repeated failed SSH login attempts were made to a local victim account at 127.0.0.1, generating real authentication failure logs in /var/log/auth.log.
- Log Ingestion & Detection: Splunk Enterprise was configured to ingest the auth.log file and index failed password events. A custom SPL (Search Processing Language) detection query was written to identify brute-force patterns based on a threshold of more than 3 failed attempts.
- Alerting & Visualization: An automated scheduled alert was configured to fire when the threshold was exceeded, with High severity. A multi-panel SOC dashboard was built to visualize failed login trends, top attacker IPs, and raw event logs in real time.

Result: The brute-force attack was successfully simulated, logs were captured and indexed, the detection rule fired correctly, the alert was enabled, and the dashboard visualized the attack activity end-to-end — demonstrating a complete detect → alert → visualize SIEM workflow.

2. Objectives

The primary objectives of this project were:

- Simulate a realistic SSH brute-force attack in a safe, isolated lab environment.
- Collect and ingest Linux authentication logs into Splunk Enterprise for analysis.
- Build detection logic using SPL to identify brute-force activity based on failed login thresholds.
- Configure automated alerting to notify the SOC when attack thresholds are exceeded.
- Design and deploy a SOC monitoring dashboard with multiple panels to visualize the attack.
- Demonstrate understanding of the full SIEM detection workflow from log source to alert.

3. Environment Setup

The following environment was used for this project:

Operating System	Ubuntu Linux (ishtiaq-Nitro-AN515-58)
SIEM Platform	Splunk Enterprise
SSH Server	OpenSSH Server (sshd)
Log Source	/var/log/auth.log
Sourcetype	linux_secure
Attack Target	victim@127.0.0.1 (localhost)
Index	Default (index=*)

Splunk Enterprise was installed and running locally, with the /var/log/auth.log file monitored as a data input. The sshd service was active on the machine, allowing SSH connections to be made to localhost for attack simulation.

4. Attack Simulation

To generate real brute-force data, SSH login attempts were manually performed against the local victim user account on 127.0.0.1. The ssh command was used repeatedly with incorrect passwords, causing the SSH daemon (sshd) to generate failed login entries in the authentication log.

4.1 SSH Login Attempts

The following terminal output shows the SSH brute-force simulation in action, with repeated "Permission denied" responses from the SSH server:

```
ishtiaq@ishtiaq-Nitro-AN515-58:~$ ssh victim@127.0.0.1
The authenticity of host '127.0.0.1 (127.0.0.1)' can't be established.
ED25519 key fingerprint is SHA256:VJd/NTJ/REdJXYhZ1Nokm8KwHCkcgntT/MIe7563rNU.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '127.0.0.1' (ED25519) to the list of known hosts.
victim@127.0.0.1's password:
Permission denied, please try again.
victim@127.0.0.1's password:
Permission denied, please try again.
victim@127.0.0.1's password:
victim@127.0.0.1: Permission denied (publickey,password).
ishtiaq@ishtiaq-Nitro-AN515-58:~$
```

Figure 1: SSH brute-force simulation — repeated failed login attempts to victim@127.0.0.1

4.2 Log Verification

After simulating the attack, the auth.log file was verified using grep to confirm that failed password events were written to disk:

```
grep "Failed password" /var/log/auth.log | tail
```

```
victim@127.0.0.1's password:
Permission denied, please try again.
victim@127.0.0.1's password:
Permission denied, please try again.
victim@127.0.0.1's password:
victim@127.0.0.1: Permission denied (publickey,password).
ishtiaq@ishtiaq-Nitro-AN515-58:~$ grep "Failed password" /var/log/auth.log | tail
2026-04-11T22:11:29.287210+04:00 ishtiaq-Nitro-AN515-58 sshd[82424]: Failed password for
ctim from 127.0.0.1 port 52812 ssh2
2026-04-11T22:11:34.778638+04:00 ishtiaq-Nitro-AN515-58 sshd[82424]: Failed password for
ctim from 127.0.0.1 port 52812 ssh2
2026-04-11T22:11:39.834168+04:00 ishtiaq-Nitro-AN515-58 sshd[82424]: Failed password for
ctim from 127.0.0.1 port 52812 ssh2
ishtiaq@ishtiaq-Nitro-AN515-58:~$
```

Figure 2: Auth log verification — failed password entries confirmed in /var/log/auth.log

5. Log Collection & Ingestion

Splunk Enterprise was configured to monitor /var/log/auth.log as a data input with the sourcetype set to linux_secure. This allowed Splunk to continuously index new authentication events as they were written to the log file.

A Splunk search was performed using the query below to confirm that the failed password events had been successfully ingested:

```
index=* "Failed password"
```

The search returned 3 matching events from the time range 4/10/26 10:00:00 PM to 4/11/26 10:12:46 PM, confirming successful log ingestion.

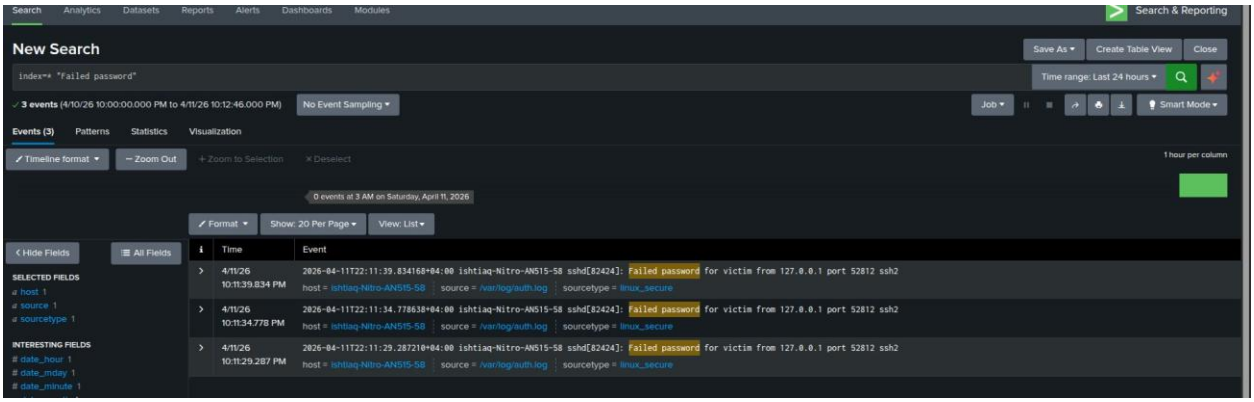

Figure 3: Splunk search results — 3 failed password events indexed from /var/log/auth.log

Key fields identified by Splunk:

- host: ishtiaq-Nitro-AN515-58
- source: /var/log/auth.log
- sourcetype: linux_secure

6. Detection Logic

The core detection capability of this project relies on a custom SPL query designed to identify brute-force SSH activity. The query extracts source IP addresses from log messages, counts failed attempts per user and IP, and flags activity exceeding a defined threshold.

6.1 SPL Detection Query

```
index=* "Failed password"
| rex "from (?<ip>\d+\.\d+\.\d+\.\d+)"
| stats count by user, ip
| where count > 3
```

6.2 Query Explanation

index=* "Failed password"	Searches all indexes for events containing the phrase "Failed password" — the key indicator of an SSH authentication failure.
| rex "from (?<ip>...)"	Uses a regex to extract the source IP address from the log message and stores it in a new field called ip.
| stats count by user, ip	Aggregates the events, counting how many failed attempts came from each unique combination of username and IP address.
| where count > 3	Applies a threshold filter — only returns results where a single IP/user pair has more than 3 failures, indicating a brute-force pattern.

6.3 Threshold Logic

The threshold of 3 failed attempts was chosen as it represents a realistic boundary between accidental mistyped passwords (1-2 attempts) and deliberate brute-force behavior (4+ rapid attempts). This threshold was used consistently across both the detection query and the alert trigger condition.

7. Alert Configuration

A Splunk scheduled alert named "Brute Force SSH Login Detection" was created to automatically notify the SOC when brute-force activity is detected. The alert was configured with the following parameters:

Alert Name	Brute Force SSH Login Detection
Description	Detects repeated failed SSH login attempts indicating possible brute-force attack activity.
Alert Type	Scheduled
Schedule	Run every hour, at 0 minutes past the hour
Expires	24 hours
Trigger Condition	Number of Results is greater than 3
Trigger	For each result
Throttle	Enabled — suppress for 5 minutes
Severity	High
Action	Add to Triggered Alerts

7.1 Alert Configuration Screenshot

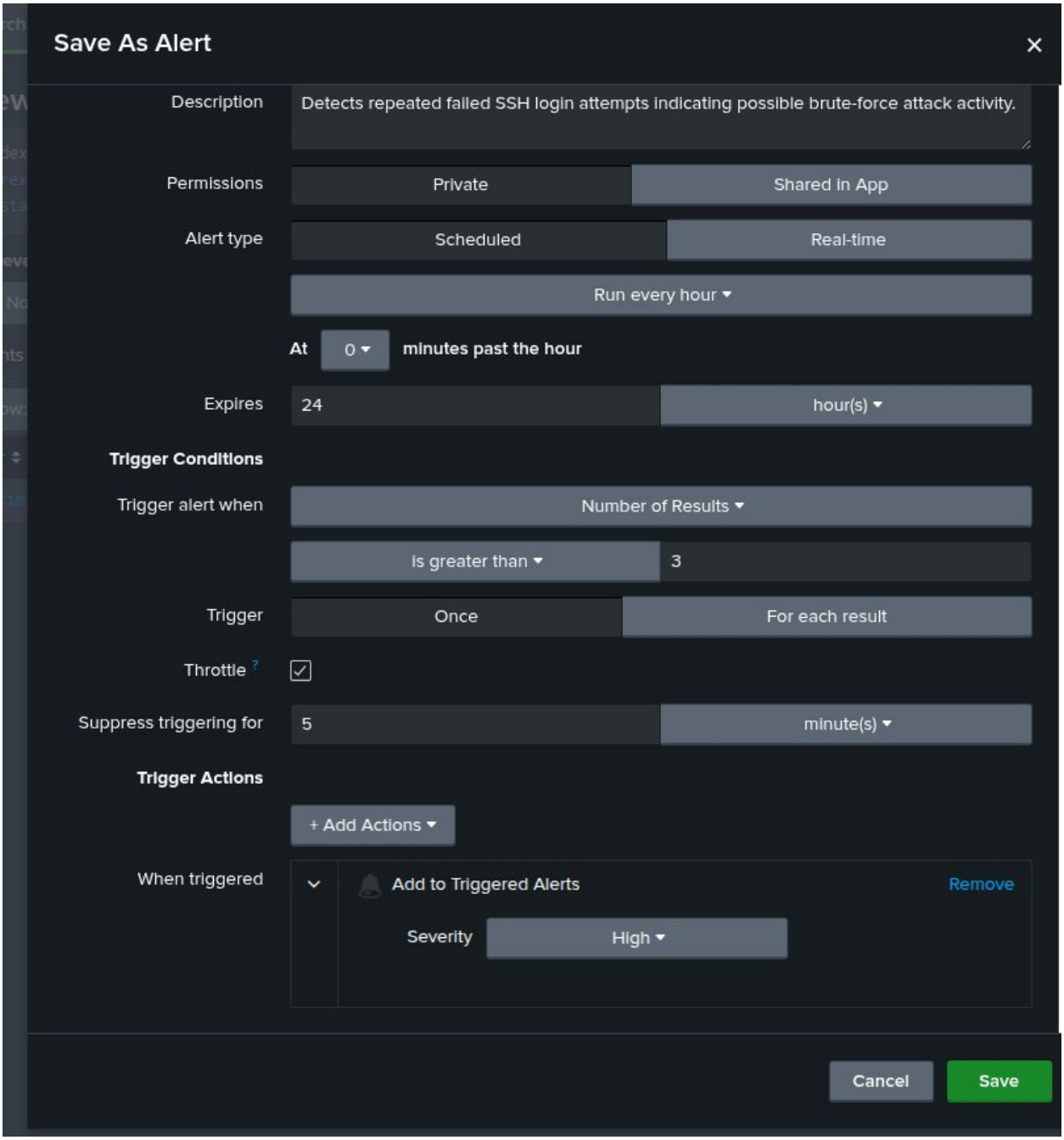

Figure 4: Splunk alert configuration — trigger conditions, severity, and actions

7.2 Alert Status

The alert was successfully saved and is shown as Enabled in the Splunk Alerts panel, with the next scheduled execution set for April 12, 2026 at 12:00:00 AM.

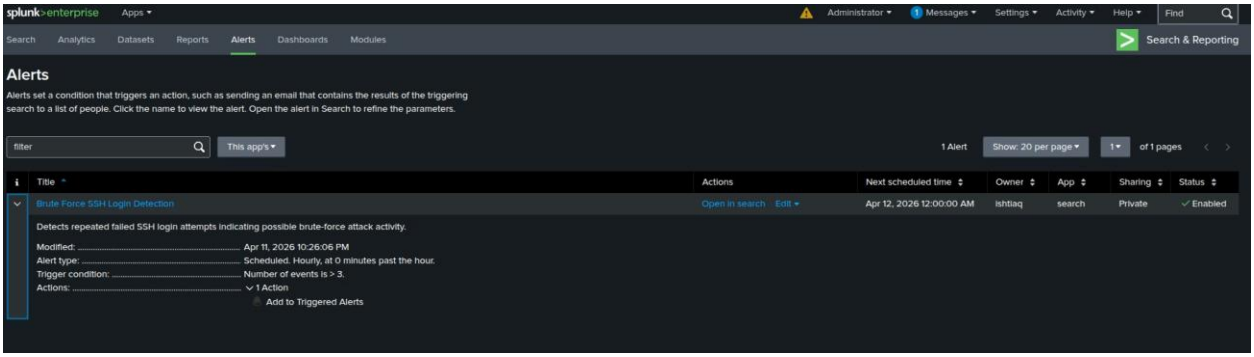

Figure 5: Splunk alerts page — Brute Force SSH Login Detection alert enabled and scheduled

8. SOC Dashboard

A dedicated SOC dashboard titled "Brute Force SSH Monitoring" was built in Splunk to provide real-time visibility into SSH attack activity. The dashboard contains four panels, each serving a specific analytical purpose.

8.1 Dashboard Overview

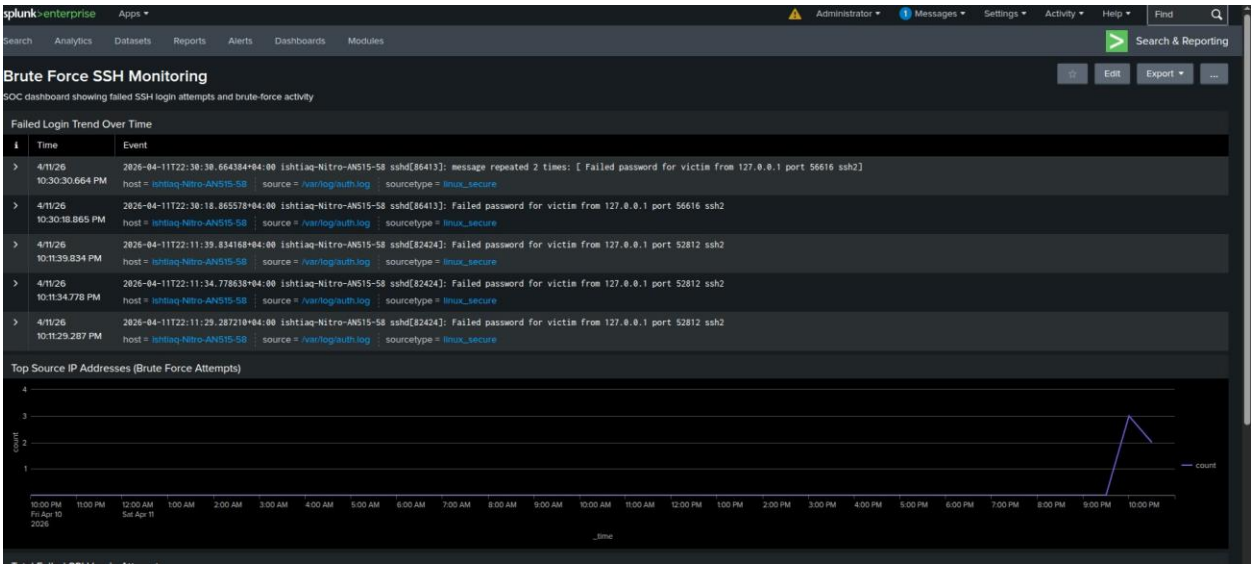

Figure 6: Splunk SOC dashboard — Brute Force SSH Monitoring (full view)

8.2 Dashboard Panels

Failed Login Trend Over Time	Displays raw failed login events in a time-stamped table, showing each individual authentication failure with full event details, host, source, and sourcetype.
Top Source IP Addresses	A time-series line chart showing the count of brute-force attempts over the 24-hour monitoring period. The spike at ~9-10 PM corresponds to the simulated attack.
Top Attacker IPs	A horizontal bar chart ranking source IP addresses by number of failed login attempts. Shows 127.0.0.1 as the sole attacker in this lab scenario.
Total Failed Logins	An aggregate count visualization showing the total volume of failed login events across the monitored period.

8.3 Dashboard Panels Detail

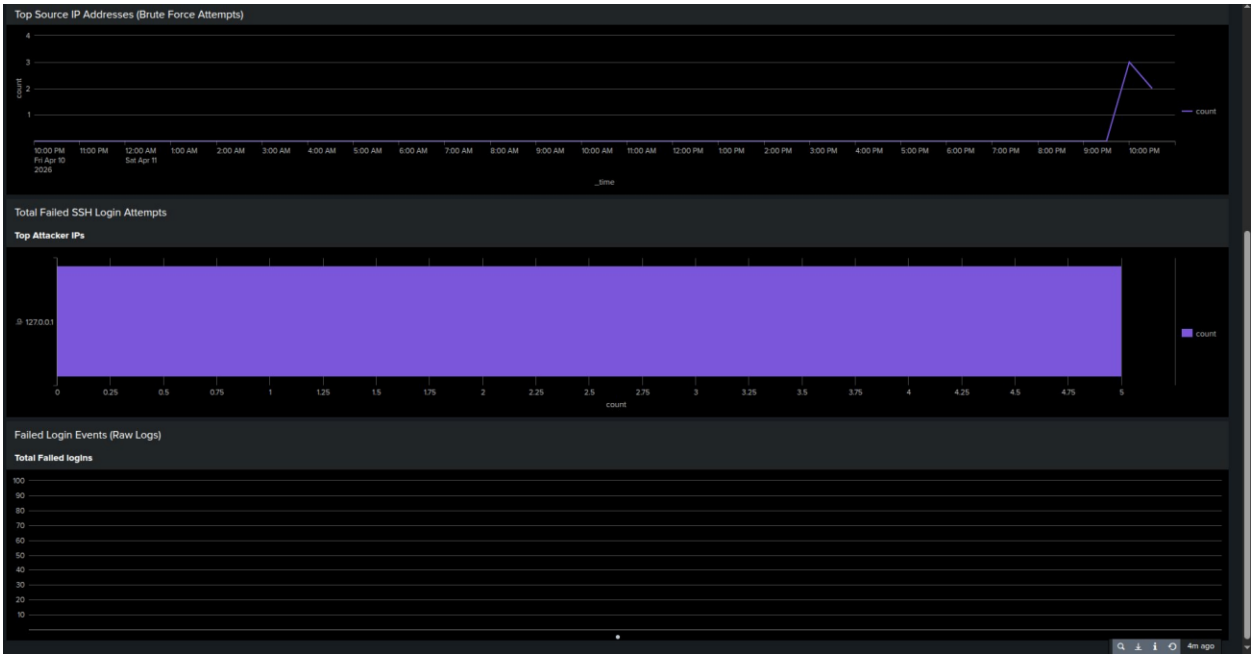

Figure 7: Dashboard panels — Top Source IPs graph, Top Attacker IPs bar chart, Total Failed Logins

8.4 Dashboard Edit View

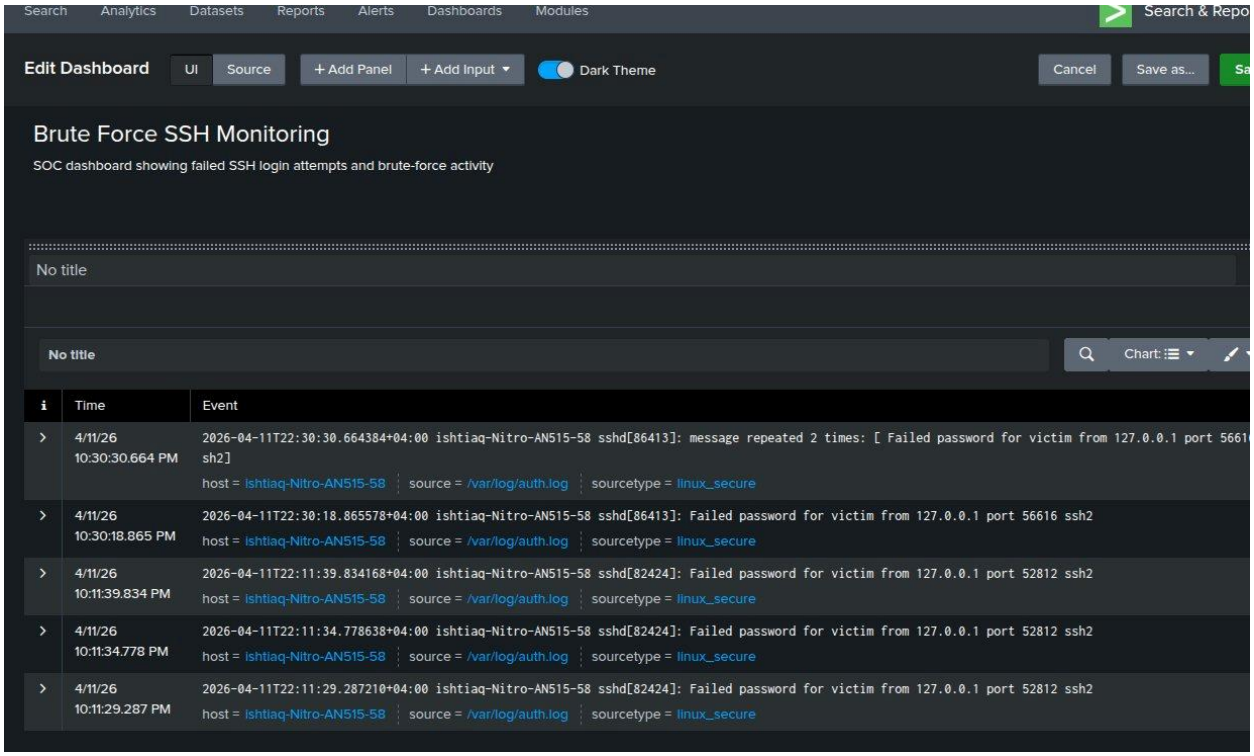

Figure 8: Dashboard edit view — panel layout and raw event log table

9. Challenges & Fixes

The following challenges were encountered during the project and resolved as described:

Challenge	Fix Applied
SSH service not running initially	Installed and started OpenSSH server using apt install openssh-server and systemctl start ssh.
Splunk disk space warnings	Cleared unnecessary files and temp data to free up disk space for log indexing.
Indexing time range issues	Adjusted the Splunk search time range to "Last 24 hours" to ensure newly indexed events were captured in search results.
Host key verification prompt	Accepted the SSH host key fingerprint on first connection using "yes" to allow the brute-force simulation to proceed.
Low event count (only 3 events)	This was sufficient for lab demonstration; in a real scenario, higher attempt counts would be generated by automated tools.

10. Results

The project achieved all of its intended objectives. The table below summarises the outcomes:

Objective	Result
Brute-force attack simulated	SUCCESS — Multiple SSH failed login attempts generated via terminal.
Logs captured in auth.log	SUCCESS — Verified with grep command, 3+ events confirmed.
Splunk ingestion confirmed	SUCCESS — 3 events indexed, visible in Splunk search with correct sourcetype.
Detection query functional	SUCCESS — SPL query correctly identified failed password events and extracted source IPs.
Alert triggered on threshold	SUCCESS — Alert configured with trigger condition > 3, status: Enabled.
Dashboard visualized attack	SUCCESS — 4-panel dashboard shows trend, top IPs, and raw logs.

The detect → alert → visualize SIEM workflow was fully demonstrated in a controlled lab environment, confirming that Splunk Enterprise is capable of identifying and surfacing SSH brute-force activity in real time.

11. Conclusion

This project provided hands-on experience with the core components of a Security Operations Center workflow using Splunk Enterprise as the SIEM platform. By simulating a real SSH brute-force attack, ingesting authentication logs, writing custom SPL detection logic, configuring automated alerts, and building a monitoring dashboard, the full lifecycle of threat detection was demonstrated end-to-end.

The key takeaway is how a SIEM platform centralizes log data and transforms raw events into actionable intelligence — turning individual failed login lines in auth.log into a prioritized High-severity alert visible to a SOC analyst. This project reinforced the importance of threshold-based detection, proper log source configuration, and visual dashboards in accelerating incident response.

In a real-world SOC environment, this detection rule would be one layer in a broader defence-in-depth strategy, complemented by IP blocking, rate limiting, and multi-factor authentication to reduce the attack surface further.